

112A Introduction to Computer and Computer Science

Homework Assignment #2

Due: 10/2 12:00:00

In this homework assignment, you will practice the basic concepts of input, branching, conditional, and iterations. As you have learned all the related knowledge during the lectures (if, while, for, exhaustive search, and so on...).

Problem #1: Taylor series expansion (50%)

Please write a python program to estimate the value of $\frac{\sin(\pi)}{\pi}$ by Taylor series expansion. The estimation error of your program should be less than 10^{-4} . The definition of estimation error, E , is

$$E = \left| \text{your estimation} - \frac{\sin(\pi)}{\pi} \right|$$

Here is the sample code,

```
print("Problem 1: Taylor series expansion")

# 自己想
# pi = 3.14159265359
# initial_guess = 1.

print("Estimation is {:.17f}".format(estimate))
```

Attention!

Importing any additional module or package is strictly prohibited in this homework.

Problem #2: Cube root (50%)

We have already illustrated how to find a **cube root** via exhaustive enumeration during the class, yet we never demonstrate how to **find a cube root via bisection search or Newton-Raphson method**. It is time for you to complete our unaccomplished duty. Please write a python program that asks users to enter one arbitrary number and print out the corresponding cube root based on the concepts of bisection search and Newton-Raphson method. You should also print out how much iterations (guesses) your program takes for searching the solution based on these two methods as well.

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
"""
Created on Sun Aug 7 01:23:45 2023

@author: Xi Winnie, student ID

@collaborators: EE Tsai, her student ID
"""
```

Please save your homework as a “.py” file, where the file name should follow this format:

112A_hw2_ID.py

For example,

112A_hw2_0123456789.py

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the “E3@NYCU” system. There will be a section for uploading your homework.